No finite C*-Smale mean-value constant at degree three

Research proof and Lean formalization

October 27, 2026

Abstract

Over the single commutative C*-algebra $C([0,1],\mathbb{C})$, no finite constant controls the Smale norm divided difference for all split monic cubics. For every proposed constant, a continuous cubic family has an invertible derivative at the test element and exactly two existing critical elements, both violating the bound. This uniform impossibility statement is stronger than a fixed counterexample to the constant-one conjecture.

Theorem 1. *Let $\mathcal{A} = C([0,1],\mathbb{C})$ with its supremum norm. For every $C \geq 0$ there are $a_1, a_2, a_3, z \in \mathcal{A}$ such that the monic cubic $P(w) = \prod_{j=1}^{3}(w - a_j)$ has $P'(z)$ invertible and exactly two distinct critical elements. Every critical element $b \in \mathcal{A}$ satisfies*

$$\frac{\|P(z) - P(b)\|}{\|z - b\|} > C \left\|P'(z)\right\|. \tag{1}$$

Here P' is the literal sum of products obtained by deleting one factor.

Proof. Write $y(t) = 2t - 1$. For $0 < \varepsilon \leq 1/16$ put

$$q_\varepsilon(t) = \sqrt{y(t)^2 + \varepsilon}, \qquad P_\varepsilon(w) = w(w - \sqrt{3}q_\varepsilon)(w + \sqrt{3}q_\varepsilon) = w^3 - 3q_\varepsilon^2 w, \qquad z = y.$$

Then

$$P'_\varepsilon(w) = 3(w - q_\varepsilon)(w + q_\varepsilon), \qquad P'_\varepsilon(y) = -3\varepsilon\, 1. \tag{2}$$

The latter is invertible. Both q_ε and $-q_\varepsilon$ are critical. Conversely, scalar evaluation of $P'_\varepsilon(b) = 0$ shows $b(t) = \pm q_\varepsilon(t)$ at each point. The quotient b/q_ε is continuous into $\{-1, 1\}$, since $q_\varepsilon > 0$. Connectedness forces one global sign, so these are exactly the two distinct critical elements. No cancellation in a function ring is being assumed.

For either critical element,

$$0 < \|y - b\| \leq \|y\| + \|q_\varepsilon\| \leq \frac{9}{4}, \qquad \left\|P'_\varepsilon(y)\right\| = 3\varepsilon. \tag{3}$$

The denominator is nonzero because $y = b$ would contradict (2). With $r = \sqrt{1 + \varepsilon}$, evaluation at the opposite endpoints gives

$$(P_\varepsilon(y) - P_\varepsilon(q_\varepsilon))(0) = 2 + 3\varepsilon + 2r^3 > 2,$$

$$(P_\varepsilon(y) - P_\varepsilon(-q_\varepsilon))(1) = -(2 + 3\varepsilon + 2r^3).$$

Consequently $\|P_\varepsilon(y) - P_\varepsilon(b)\| > 2$ for every critical element, uniformly in ε.

Given $C \geq 0$, choose

$$\varepsilon = \frac{1}{16(C+1)}. \tag{4}$$

This is strictly positive and at most $1/16$, and

$$C\left\|P'_\varepsilon(y)\right\|\|y-b\| \le \frac{27}{4}C\varepsilon = \frac{27C}{64(C+1)} \le \frac{27}{64} < 2.$$

Together with the value gap this proves (1). □

Relation to the source and prior refutations. Krishna [1, Theorems 1.1–1.2] recalls the scalar Smale theorem with universal constant 4, and Conjecture 2.1 proposes a C*-algebraic constant-one extension. Public scalar counterexamples, including [2, Theorem 1.3], already refute that universal conjecture because $\mathbb{C}$ is itself a commutative C*-algebra. We do not claim its first resolution. Theorem 1 rules out every finite constant at the fixed degree three over one fixed function algebra; the scalar constant 4 remains valid. The source's degree-two theorem is known and receives no credit here.

Formal scope and review status. The Lean certificate has 49 declarations in five modules, including four reused continuous-selection lemmas. The closed endpoint

`CStarSmaleUnbounded.infinite_cubic_constant`

contains the universal quantifier over C, the actual monic degree-three product, invertibility of the test derivative, exact two-element critical set and strict norm violation at every critical element. The second endpoint `no_finite_cubic_bound` negates the existence of a nonnegative universal cubic bound. All source product/derivative and norm bridges, (4) and its positivity are proved. There is no residual small- parameter or selection hypothesis.

The isolated strict replay passes under Lean 4.34.1 and mathlib revision

`d13f23b723b8a846827a245b89c10fc7d3f11612`,

using only `propext`, `Classical.choice` and `Quot.sound`. Independent final source, mathematical, formal, manuscript and bounded novelty review passes. Eligibility is solely as a new infinite-class theorem, not credit for an unresolved source question. Acceptance also requires a fresh replay of the extracted proof archive. The rejected fixed constant-one witness is not counted; the parameter family is one combined outcome. No worldwide priority claim is made.

References

[1] K. Mahesh Krishna, *C*-algebraic Smale mean value conjecture and Dubinin–Sugawa dual mean value conjecture*, Vestnik KRAUNC. Fiz.-Mat. Nauki **54**(2) (2026), 38–47. doi:10.26117/2079-6641-2026-54-2-38-47.

[2] A. Jatar and T.-W. Ng, *Smale's mean value constant and admissible univalent Blaschke pairs*, arXiv:2610.21472v2 (2026).